

(c) Population and sampling		
1c.01	Know the difference between population, sample frame and sample.	Identify a population, and suggest a suitable sampling frame.
1c.02	Know that 'population' can have different meanings within a stated context.	For example, all employees in an office; all females in the UK; all items produced in a factory.
1c.03	Know reasons for employing judgement sampling or opportunity (convenience) sampling, and the associated risks of bias when these techniques are used.	Including use of cluster sampling <u>and quota sampling</u> . Reasons including factors such as convenience, cost and time.
1c.04	a. Know appropriate sampling techniques in the context of the problem to avoid bias. b. Understand random, systematic, and <u>quota</u> sampling.	Including advantages and disadvantages of each technique. e.g. Know that systematic <u>and quota</u> sampling techniques are generally non-random. Know that the period of systematic sampling may coincide with a period occurring in the data.
1c.05	Know the key features of a simple random sample <u>and demonstrate understanding of how different techniques, both physical and electronic, are used to select random members from a population: including, but not limited to, dice, cards, random number lists, and calculator functions.</u>	Be aware that all items in the population should have the same likelihood of inclusion in a simple random sample. <u>Selection of items for a sample may be required, including dealing with issues such as repeated random numbers and random numbers out of range.</u>
1c.06	<u>Use stratification and know when this is appropriate before sampling takes place.</u>	Identify suitable strata, e.g. gender or age group. <u>Including the calculation of appropriate strata sizes. Stratifying by one or more than one category.</u>